

Lab 11

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Roll no : 38

Submitted to : sir Shoaib

Section : A

Task 0:

```
gh codespace create --repo 23-22412-038-sys/Lab11
✓ Codespaces usage for this repository is paid for by 23-22412-038-sys
error getting devcontainer.json paths: HTTP 400: The 'ref' provided was invalid.
Please specify a valid branch name or commit SHA (https://api.github.com/repos/1125688696/codespaces/devcontainers?per_page=100&ref=main)

Alli@DESKTOP-KQE4P8D MINGW64 ~ (main)
$ gh codespace list


| NAME           | DISPLAY NAME   | REPOSITORY     | BRANCH | STATE    | CREATED AT     |
|----------------|----------------|----------------|--------|----------|----------------|
| sturdy-para... | sturdy para... | 23-22412-03... | main   | Shutdown | about 16 da... |
| obscure-zeb... | obscure zebra  | 23-22412-03... | main*  | Shutdown | about 16 da... |
| reimagined-... | reimagined ... | 23-22412-03... | main   | Shutdown | about 2 day... |
| ominous-waf... | ominous waffle | 23-22412-03... | main*  | Shutdown | about 2 day... |



Alli@DESKTOP-KQE4P8D MINGW64 ~ (main)
$ gh repo delete 23-22412-038-sys/Lab11 --yes
HTTP 403: Must have admin rights to Repository. (https://api.github.com/repos/23-22412-038-sys/Lab11)
This API operation needs the "delete_repo" scope. To request it, run: gh auth refresh -h github.com -s delete_repo

Alli@DESKTOP-KQE4P8D MINGW64 ~ (main)
$ gh repo create Lab11 --public --add-readme
HTTP 422: Repository creation failed. (https://api.github.com/user/repos)
? Choose codespace: 23-22412-038-sys/Lab11 [main]: didactic invention
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation: https://help.ubuntu.com
 * Management:   https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

23-22412-038-sys → /workspaces/Lab11 (main) $
```

Task 1

```
23-22412-038-sys → /workspaces/Lab11 (main) $
23-22412-038-sys → /workspaces/Lab11 (main) $ touch main.tf
23-22412-038-sys → /workspaces/Lab11 (main) $
```

```
codepaces@codepaces-79c5t5: nano/main$
GNU nano 2.9.3
provider "aws" {
  region                  = "us-central-1" # Middle East (UAE)
  shared_config_files     = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
```

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13:17 PM 12/30/2025

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
codepaces@codepaces-79c51b: nano main.tf
GNU nano 2.10
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
  type = string
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

File Name to write: main.tf

Help Cancel

Doc Format Mac Format

Append Prepend

Backup File Browse

11:29 PM 12/30/2025

```
#23-27412-038-sys ~ /workspaces/Lab11 (main) $ terraform apply -auto-approve
var.subnet_cidr_block
  Enter a value: 1

Changes to Outputs:
  + subnet_cidr_block_output = "1"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "1"
```

```
codepaces@codepaces-79c51b: nano main.tf
GNU nano 2.2
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
  type = string
  default = "10.0.0.0/24"
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

```
R23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  - subnet_cidr_block_output = "1" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.0.0/24"
```

```
R23-22412-038-sys + /workspaces/Lab11 (main) $ export TF_VAR_subnet_cidr_block=10.0.20.0/24
terraform apply -auto-approve

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.20.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.20.0/24"
```

```
R23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.20.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.30.0/24"
```

```
R23-22412-038-sys + /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.40.0/24"

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.40.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.40.0/24"
```

```
823-22412-038-sys → /workspaces/Lab11 (main) $ printenv | grep TF_VAR_
unset TF_VAR_subnet_cidr_block
printenv | grep TF_VAR_
TF_VAR_subnet_cidr_block=10.0.20.0/24
```

Task 2

```
823-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.0.0/24"

Changes to Outputs:
  - subnet_cidr_block_output = "10.0.40.0/24" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.0.0/24"
```

```
823-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token_12345"

Changes to Outputs:
  + api_session_token_output = (sensitive value)
  - subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.30.0/24"
```

```
terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 7,
  "lineage": "0d7ac30c-597d-354f-956a-9b5b08f74f35",
  "outputs": {
    "api_session_token_output": {
      "value": "my_API_session_Token_12345",
      "type": "string",
      "sensitive": true
    },
    "subnet_cidr_block_output": {
      "value": "10.0.30.0/24",
      "type": "string"
    }
  },
  "resources": [],
  "check_results": [
    {
      "object_kind": "var",
      "config_addr": "var.api_session_token",
      "status": "pass",
      "objects": [
        {
          "object_addr": "var.api_session_token",
          "status": "pass"
        }
      ]
    },
    {
      "object_kind": "var",
      "config_addr": "var.subnet_cidr_block",
      "status": "pass",
      "objects": [
        {
          "object_addr": "var.subnet_cidr_block",
          "status": "pass"
        }
      ]
    }
  ]
}
```

```
823-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token_12345"

Changes to Outputs:
  + api_session_token_output = (sensitive value)
  - subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
```

```
823-22412-038-sys → /workspaces/Lab11 (main) $ cat terraform.tfstate | grep api_session_token -n
7:   "api_session_token_output": {
21:     "config_addr": "var.api_session_token",
25:     "object_addr": "var.api_session_token",
```

Task 3


```
code@ec2-10-0-30-0:~/workspace-79c51b: nano main.tf
GNU nano 2.9.3
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

variable "subnet_cidr_block" {
  type = string
  default = ""
  description = "CIDR block to assign to the application subnet"
  sensitive = false
  nullable = false
  ephemeral = false

  validation {
    condition = can(regex("^[0-9]{1,3}\\.[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string."
  }
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

variable "api_session_token" {
  type = string
  default = ""
  sensitive = true
  nullable = false
  ephemeral = false

  validation {
    condition = can(regex("^[A-Za-z0-9-]{20,}$", var.api_session_token))
    error_message = "Token must be 20+ characters."
  }
}

output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}

variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ aws ec2 describe-subnets \
--filters "Name=availability-zone,Values=me-central-1a" \
--query "Subnets[].SubnetId" \
--output text
subnet-08a73f8aa1dee72d3
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat terraform.tfvars
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-08a73f8aa1dee72d3"
"
subnet_count      = 3
monitoring        = true
@23-22412-038-sys → /workspaces/Lab11 (main) $
```

```
codepaces@codepaces-79c31b: nano locals.tf
GNU nano 2.2 locals.tf
locals {
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count = var.subnet_count
  is_production = var.environment == "prod"
  monitoring_enabled = var.monitoring || !local.is_production
}
```

```
codepaces@codepaces-79c31b: nano main.tf
GNU nano 2.2 main.tf
value = var.subnet_cidr_block
}
variable "api_session_token" {
  type = string
  default = ""
  sensitive = true
  nullable = false
  ephemeral = false
  validation {
    condition = can(regex("^[A-Za-z0-9-]{20,}$", var.api_session_token))
    error_message = "Token must be 20+ characters."
  }
}
output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
output "resource_name" {
  value = local.resource_name
}
output "primary_public_subnet" {
  value = local.primary_public_subnet
}
output "subnet_count" {
  value = local.subnet_count
}
output "is_production" {
  value = local.is_production
}
output "monitoring_enabled" {
  value = local.monitoring_enabled
}
```

```

023-22412-038-sys ➔ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  - api_session_token_output = (sensitive value)
  + is_production            = false
  + monitoring_enabled       = true
  + primary_public_subnet    = "subnet-08a73f8aa1dee72d3"
  + resource_name            = "lab_work-dev"
  - subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.0.0/24"
  + subnet_count             = 3

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3

```

Task 4

```

023-22412-038-sys ➔ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + tags = {
    + Environment = "dev"
    + Owner       = "platform-team"
    + Project     = "sample-app"
  }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner"       = "platform-team"
  "Project"     = "sample-app"
})

```

```

023-22412-038-sys ➔ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + server_config = {
    + backup_enabled = false
    + instance_type  = "t3.micro"
    + monitoring     = true
    + name           = "web-server"
    + storage_gb     = 20
  }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type"  = "t3.micro"
  "monitoring"     = true
  "name"           = "web-server"
  "storage_gb"     = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner"       = "platform-team"
  "Project"     = "sample-app"
})
023-22412-038-sys ➔ /workspaces/Lab11 (main) $

```

Task 5


```
codepaces@codepaces-79c15b: nano main.tf
GNU nano 2.2
variable "server_config" {
  type = object({
    name           = string
    instance_type  = string
    monitoring      = bool
    storage_gb     = number
    backup_enabled = bool
  })
  description = "Server configuration object"
}

output "server_config" {
  value = var.server_config
}

variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
  description = "List of server names (duplicates allowed)"
}

variable "server_metadata" {
  type = tuple([string, number, bool])
  default = ["web-1", 4, true]
  description = "Tuple of server info (immutable, mixed types)"
}

variable "availability_zones" {
  type = set(string)
  default = ["us-central-1a", "us-central-1b"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}
```

Outputs:

```
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "us-central-1a",
    "us-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})

```

```
codepace@codepace-79c31b: nano locals.tf
GNU nano 2.2
locals {
  # Task 2: locals
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count = var.subnet_count
  is_production = var.environment == "prod"
  monitoring_enabled = var.monitoring || local.is_production

  # Task 3: mutation Locals
  mutated_list = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-2"])
  mutated_set = setunion(var.availability_zones, ["us-central-1c"])
}

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```

```
codepace@codepace-79c31b: nano main.tf
GNU nano 2.2
main.tf
name = string
instance_type = string
monitoring = bool
storage_gb = number
backup_enabled = bool
}
description = "Server configuration object"
}

output "server_config" {
  value = var.server_config
}

variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
  description = "List of server names (duplicates allowed)"
}

variable "server_metadata" {
  type = tuple([string, number, bool])
  default = ["web-1", 4, true]
  description = "Tuple of server info (immutable, mixed types)"
}

variable "availability_zones" {
  type = set(string)
  default = ["us-central-1b", "us-central-1a", "us-central-1c"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple = local.mutated_tuple
  }
}

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```

```
#23-22412-036-sys ~ /workspaces/Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
+ mutation_comparison = {  
  + mutated_tuple = [  
    + "4",  
    + "true",  
    + "web-1",  
    + "web-2",  
  ]  
  + original_tuple = [  
    + "web-1",  
    + 4,  
    + true,  
  ]  
}
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>  
compare_collections = {  
  "list_example" = tolist([  
    "web-2",  
    "web-1",  
    "web-2",  
  ])  
  "set_example" = toset([  
    "me-central-1a",  
    "me-central-1b",  
  ])  
  "tuple_example" = [  
    "web-1",  
    4,  
    true,  
  ]  
}  
is_production = false  
monitoring_enabled = true  
mutation_comparison = {  
  "mutated_tuple" = toset([  
    "4",  
    "true",  
    "web-1",  
    "web-2",  
  ])  
  "original_tuple" = [  
    "web-1",  
    4,  
    true,  
  ]  
}
```

```

    })
    "set_example" = toset([
        "me-central-1a",
        "me-central-1b",
    ])
    "tuple_example" = [
        "web-1",
        4,
        true,
    ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
    "mutated_tuple" = toset([
        "4",
        "true",
        "web-1",
        "web-2",
    ])
    "original_tuple" = [
        "web-1",
        4,
        true,
    ]
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
    "backup_enabled" = false
    "instance_type" = "t3.micro"
    "monitoring" = true
    "name" = "web-server"
    "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
    "Environment" = "dev"
    "Owner" = "platform-team"
    "Project" = "sample-app"
})
})

```

Task 6

```
codepath@codepath:79c31b: nano locals.tf
GNU nano 2.2
locals.tf
locals {
  # Task 2: locals
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count = var.subnet_count
  is_production = var.environment == "prod"
  monitoring_enabled = var.monitoring || !local.is_production

  # Task 3: mutation locals
  mutated_list = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-2"])
  mutated_set = setunion(var.availability_zones, ["us-central-1a"])

  # Task 4: null/dynamic locals
  server_tag = merge(
    { name = "web-server" },
    var.optional_tag != null ? { Custom = var.optional_tag } : {}
  )
}

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next
```

```
codepath@codepath:79c31b: nano main.tf
GNU nano 2.2
main.tf
type = tuple(string, number, bool)
default = ["web-1", 4, true]
description = "Tuple of server info (immutable, mixed types)"

variable "availability_zones" {
  type = set(string)
  default = ["us-central-1b", "us-central-1a", "us-central-1b"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple = local.mutated_tuple
  }
}

variable "optional_tag" {
  type = string
  description = "A tag that may or may not be provided"
  default = null
}

variable "dynamic_value" {
  type = any
  description = "A variable that can accept any data type"
  default = null
}

output "optional_tag" {
  value = local.server_tags
}

output "value_received" {
  value = var.dynamic_value
}
```



```
codospac@codospac:79c37b16/vbwh
$ cd /workspaces/Lab11 (main)
$ nano main.tf
$ nano locals.tf
$ nano main.tf
$ terraform apply -auto-approve

Changes to Outputs:
  optional_tag = [
    "web-server"
  ]

you can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = -sensitive
compare_collections = [
  "list_example" = tolist([
    "web-1",
    "web-1",
    "web-2"
  ])
  "set_example" = toset([
    "ne-central-1a",
    "ne-central-1b"
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
]
is_production = false
monitoring_enabled = true
mutation_comparison = [
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2"
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
]
}
```

```
codospac@codospac:79c37b16/vbwh
$ cd /workspaces/Lab11 (main)
$ nano main.tf
$ nano locals.tf
$ nano main.tf
$ terraform apply -auto-approve

Changes to Outputs:
  optional_tag = [
    "web-server"
  ]
primary_public_subnet = "subnet-08a73f8a5d6e72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gp" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
])
}
```

```
codepaces@codepaces-79c17b: /bin/bash
})
set_example = toset([
    "ne-central-1a",
    "ne-central-1b",
])
tuple_example = [
    "web-1",
    4,
    true,
]
}
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
    mutated_tuple = toset([
        "4",
        "true",
        "web-1",
        "web-2",
    ])
    original_tuple = [
        "web-1",
        4,
        true,
    ]
}
}
optional_tag = {
    "Custom" = "dev"
    "Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
    "backup_enabled" = false
    "instance_type" = "t3.micro"
    "monitoring" = true
    "name" = "web-server"
    "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
    "Environment" = "dev"
    "Owner" = "platform-team"
    "Project" = "sample-app"
])
value_received = "hello"
R03-27412-033-aps + /workspaces/lab11 (note) 1 |
```

```
codepaces@codepaces-79c17b: /bin/bash
})
set_example = toset([
    "ne-central-1a",
    "ne-central-1b",
])
tuple_example = [
    "web-1",
    4,
    true,
]
}
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
    mutated_tuple = toset([
        "4",
        "true",
        "web-1",
        "web-2",
    ])
    original_tuple = [
        "web-1",
        4,
        true,
    ]
}
}
optional_tag = {
    "Custom" = "dev"
    "Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
    "backup_enabled" = false
    "instance_type" = "t3.micro"
    "monitoring" = true
    "name" = "web-server"
    "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
    "Environment" = "dev"
    "Owner" = "platform-team"
    "Project" = "sample-app"
])
value_received = "42"
R03-27412-033-aps + /workspaces/lab11 (note) 1 |
```

```
codepaces@codepaces-79c17b: /bin/bash
})
tuple_example = [
  "web-1",
  4,
  true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-08a79f8a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
])
value_received = [
  "a",
  "b",
  "c",
]
023-22422-038-aps - /workspaces/Lab11 (note) $
```

```
codepaces@codepaces-79c17b: /bin/bash
"ne-central-1b",
})
tuple_example = [
  "web-1",
  4,
  true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-08a79f8a1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap([
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
])
value_received = {
  "cpu" = 4
  "name" = "server"
}
023-22422-038-aps - /workspaces/Lab11 (note) $
```

```

@23-22412-038-sys /workspaces/Lab11 (main) $ terraform apply -auto-approve
Error: Missing attribute value

on terraform.tfvars line 22:
22: dynamic_value = { "null"
23:   name = "server"
}

Expected an attribute value, introduced by an equals sign ("=").

@23-22412-038-sys /workspaces/Lab11 (main) $
@23-22412-038-sys /workspaces/Lab11 (main) $
@23-22412-038-sys /workspaces/Lab11 (main) $ terraform apply -auto-approve
@23-22412-038-sys /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  value_received = [
    ~ CPU = 4
    ~ name = "server"
  ] => null

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = (sensitive)
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
}

```

Task 7

```

@23-22412-038-sys → /workspaces/Lab11 (main) $ touch .gitignore
@23-22412-038-sys → /workspaces/Lab11 (main) $ nano .gitignore
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat .gitignore
.terraform/*
*.tfstate
*.tfstate.*
*.tfvars
*.pem
@23-22412-038-sys → /workspaces/Lab11 (main) $

```

Task 8

```

@23-22412-038-sys /workspaces/Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Saving previous version of HashiCorp/terraform from the dependency lock file.
- Using previously-installed hashicorp/terraform v0.12.0.

Terraform has been successfully initialized!

You may now begin working with Terraform. Running "terraform plan" will
show any changes that are required for your configuration. All Terraform
commands should now work.

If you ever want to change modules or override configuration for Terraform,
visit this link to learn the format: https://www.terraform.io/docs/configuration/0.12.html
Terraform uses the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ create
Terraform will perform the following actions:

# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
  arn = (known after apply)
  id = (known after apply)
  owner_id = (known after apply)
  region = "us-east-1"
  tags = {
    "Environment" = "dev"
    "Name" = "aws-igw"
    "Owner" = "terraform"
  }
  tags_all = {
    "Environment" = "dev"
    "Name" = "aws-igw"
    "Owner" = "terraform"
  }
  vpc_id = (known after apply)
}

# aws_route_table.main will be created
+ resource "aws_route_table" "main" {

```

```
codepaces@codepaces-79c1tb:~/vlab$  
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.  
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:  
+ create  
Terraform will perform the following actions:  
  
# aws_internet_gateway.igw will be created  
+ resource "aws_internet_gateway" "igw" {  
  + arn = (known after apply)  
  + id = (known after apply)  
  + owner_id = (known after apply)  
  + region = "us-central-1"  
  + tags = {  
    + "Environment" = "dev"  
    + "Name" = "myproject-igw"  
    + "Owner" = "toshal"  
  }  
  + tags_all = {  
    + "Environment" = "dev"  
    + "Name" = "myproject-igw"  
    + "Owner" = "toshal"  
  }  
  + vpc_id = (known after apply)  
}  
  
# aws_route_table.main_rt will be created  
+ resource "aws_route_table" "main_rt" {  
  + arn = (known after apply)  
  + id = (known after apply)  
  + owner_id = (known after apply)  
  + propagating_vgws = (known after apply)  
  + region = "us-central-1"  
  + route = [  
    + {  
      + cidr_block = "0.0.0.0/0"  
      + gateway_id = (known after apply)  
      + (11 unchanged attributes hidden)  
    }  
  ]  
  + tags = {  
    + "Environment" = "dev"  
    + "Name" = "myproject-rt"  
    + "Owner" = "toshal"  
  }  
  + tags_all = {  
    + "Environment" = "dev"  
  }  
}
```

```
codepaces@codepaces-79c1tb:~/vlab$  
aws_route_table_association.subnet_assoc: Creating...  
aws_route_table_association.subnet_assoc: Creation complete after 1s [id=rtbassoc-06e066178d3fd835]  
Apply complete! Resources: 5 added, 0 changed, 0 destroyed.  
  
Outputs:  
api_session_token_output = <sensitive>  
compare_collections = {  
  "list_example" = tolist([  
    "web-1",  
    "web-2",  
  ])  
  "set_example" = toset([  
    "us-central-1a",  
    "us-central-1b",  
  ])  
  "tuple_example" = [  
    "web-1",  
    4,  
    true,  
  ]  
}  
is_production = false  
monitoring_enabled = true  
mutation_comparison = {  
  "mutated_tuple" = toset([  
    "4",  
    "true",  
    "web-1",  
    "web-2",  
  ])  
  "original_tuple" = [  
    "web-1",  
    4,  
    true,  
  ]  
}  
optional_tag = "dev"  
primary_public_subnet = "subnet-0123456789abcdef0"  
resource_name = "myproject-dev"  
subnet_cidr_block_output = "10.0.0.0/24"  
subnet_count = 1  
value_received = {  
  "cpu" = 4  
  "name" = "server"  
}
```

Task 9

```

M3-22412-038-sys ~ /workspaces/Lab11 (main) $ terraform init
terraform apply -auto-approve
initializing the backend...
initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
aws_route_table_association.subnet_assoc: Refreshing state... [id=rtbassoc-08e08b6178d3fd835]
aws_vpc.main_vpc: Refreshing state... [id=vpc-0cd12f95de8556417]
aws_subnet.app_subnet: Refreshing state... [id=subnet-0b1696fda58ee7a71]
aws_internet_gateway.igw: Refreshing state... [id=igw-078d80b905e5f9c60]
aws_route_table.main_rt: Refreshing state... [id=rtb-079ccd3c3c57d9571]

Terraform used the selected providers to generate the following execution plan: Resource actions are indicated with the following symbols:
+ create
+ destroy

Terraform will perform the following actions:

# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
+   arn                = (known after apply)
+   default_route_table_id = (known after apply)
+   id                 = (known after apply)
+   owner_id           = (known after apply)
+   region             = "us-central-1"
}

```

```

Terraform used the selected providers to generate the following execution plan: Resource actions are indicated with the following symbols:
+ create
+ destroy

Terraform will perform the following actions:

# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
+   arn                = (known after apply)
+   default_route_table_id = (known after apply)
+   id                 = (known after apply)
+   owner_id           = (known after apply)
+   region             = "us-central-1"
+   route              = [
+     {
+       cidr_block      = "0.0.0.0/0"
+       gateway_id      = (known after apply)
+       # (10 unchanged attributes hidden)
+     },
+   ],
+   tags               = {
+     "Name" = "dev-rt"
+   }
+   tags_all           = {
+     "Name" = "dev-rt"
+   }
+   vpc_id             = (known after apply)
}

# aws_default_security_group.myapp_sg will be created
+ resource "aws_default_security_group" "myapp_sg" {
+   arn                = (known after apply)
+   description         = (known after apply)
+   egress              = [
+     {
+       cidr_blocks     = [
+         "0.0.0.0/0"
+       ]
+       from_port        = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids  = []
+       protocol         = "-1"
+       security_groups  = []
+       self              = false
+       to_port          = 0
+     },
+   ],
+   # (10 unchanged attributes hidden)
}

```



```

+ self = false
+ to_port = 0
# (1 unchanged attribute hidden)
},
]
+ id = (known after apply)
+ ingress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
+ ]
+ from_port = 80
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 80
# (1 unchanged attribute hidden)
+ },
+ {
+ cidr_blocks = [
+ "4.240.18.229/32",
+ ]
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 22
# (1 unchanged attribute hidden)
+ },
]
+ name = (known after apply)
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ region = "me-central-1"
+ revoke_rules_on_delete = false
+ tags = {
+ "Name" = "dev-sg"
+ }
+ tags_all = {
+ "Name" = "dev-sg"
+ }
+ vpc_id = (known after apply)
}

```

```

# aws_internet_gateway.myapp_igw will be created
+ resource "aws_internet_gateway" "myapp_igw" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ region = "me-central-1"
+ tags = {
+ "Name" = "dev-igw"
+ }
+ tags_all = {
+ "Name" = "dev-igw"
+ }
+ vpc_id = (known after apply)
}

# aws_key_pair.ssh_key will be created
+ resource "aws_key_pair" "ssh_key" {
+ arn = (known after apply)
+ fingerprint = (known after apply)
+ id = (known after apply)
+ key_name = "serverkey"
+ key_name_prefix = (known after apply)
+ key_pair_id = (known after apply)
+ key_type = (known after apply)
+ public_key = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPNw9F8K0FKZ8hsRwTKXLCwL0kr5yt861DI/qQ+TMD codespace@codespaces-79c519"
+ region = "me-central-1"
+ tags_all = (known after apply)
}

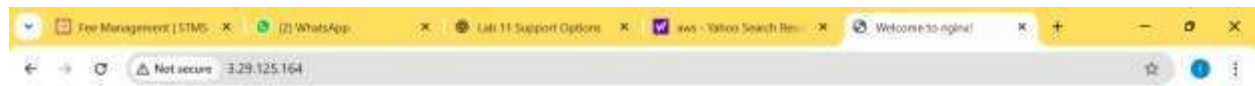
```



```
~/lib/
[ec2-user@ip-10-0-10-168 ~]$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```



Cleanup

```

003-22412-008-sys @ /workspaces/Lab11 (main) % terraform destroy -auto-approve
aws_key_pair.ssh_key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-08c0de790de26507d]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-08deab4a9bafbaac7]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-03e5bda232f8fa581]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-0a106f65daa2ea6eb]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0e0e0308f83c69ff2]
aws_instance.myapp_server: Refreshing state... [id=i-097e723716367b59c]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```

# aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn              = "arn:aws:ec2:me-central-1:623705167686:route-table/rtb-0e0e0308f83c69ff2" -> null
  - default_route_table_id = "rtb-0e0e0308f83c69ff2" -> null
  - id                  = "rtb-0e0e0308f83c69ff2" -> null
  - owner_id            = "623705167686" -> null
  - propagating_vgws     = [] -> null
  - region              = "me-central-1" -> null
  - route               = [
    - {
      - cidr_block      = "0.0.0.0/0"
      - gateway_id      = "igw-08deab4a9bafbaac7"
      # (10 unchanged attributes hidden)
    }
  ] -> null
  - tags               = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all            = {
    - "Name" = "dev-rt"
  } -> null
  - vpc_id              = "vpc-08c0de790de26507d" -> null
}

# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
  - arn              = "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0a106f65daa2ea6eb" -> null
  - description      = "default VPC security group" -> null
  - egress            = [
    - {
      - cidr_blocks      = [
        - "0.0.0.0/0",
      ]
      - from_port         = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids   = []
      - protocol           = "-1"
      - security_groups   = []
      - self               = false
      - to_port           = 0
    }
  ]
}

```

```

# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
  - arn              = "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0a106f65daa2ea6eb" -> null
  - description      = "default VPC security group" -> null
  - egress            = [
    - {
      - cidr_blocks      = [
        - "0.0.0.0/0",
      ]
      - from_port         = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids   = []
      - protocol           = "-1"
      - security_groups   = []
      - self               = false
      - to_port           = 0
    }
  ]
}

```

```

# aws_instance.myapp_server will be destroyed
- resource "aws_instance" "myapp_server" {
  - ami              = "ami-05524d6658fcf35b6" -> null
  - arn              = "arn:aws:ec2:me-central-1:623705167686:instance/i-097e723716367b59c" -> null
  - associate_public_ip_address = true -> null
  - availability_zone = "me-central-1a" -> null
  - disable_api_stop    = false -> null
  - disable_api_termination = false -> null
  - ebs_optimized        = false -> null
  - force_destroy        = false -> null
  - get_password_data    = false -> null
  - hibernation           = false -> null
  - id                   = "i-097e723716367b59c" -> null
  - instance_initiated_shutdown_behavior = "stop" -> null
  - instance_state       = "running" -> null
  - instance_type         = "t3.micro" -> null
  - ipv6_address_count    = 0 -> null
  - ipv6_addresses        = [] -> null
  - key_name              = "serverkey" -> null
  - monitoring            = false -> null
  - placement_partition_number = 0 -> null
  - primary_network_interface_id = "eni-05c432978c007a7ce" -> null
  - private_dns           = "ip-10-0-10-168.me-central-1.compute.internal" -> null
  - private_ip            = "10.0.10.168" -> null
  - public_ip             = "3.28.125.164" -> null
  - region               = "me-central-1" -> null
  - secondary_private_ips  = [] -> null
  - security_groups       = [] -> null
  - source_dest_check      = true -> null
  - subnet_id             = "subnet-03e5bda232f8fa581" -> null
  - tags                  = {
    - "Name" = "dev-ec2-instance"
  } -> null
  - tags_all              = {
    - "Name" = "dev-ec2-instance"
  } -> null
  - tenancy                = "default" -> null
  - user_data              = <--EOT
    #!/bin/bash
    yum update -y
    yum install -y nginx
    systemctl start nginx
  }
}

```



```
# aws_key_pair.ssh_key will be destroyed
resource "aws_key_pair" "ssh_key" {
  arn          = "arn:aws:ec2:me-central-1:623705167686:key-pair/serverkey" -> null
  fingerprint = "EChX8mDVY3SPkyVYNh9KakJ424WagsWqyFvd/1ztXo" -> null
  id          = "serverkey" -> null
  key_name    = "serverkey" -> null
  key_pair_id = "key-D45da68074c103868" -> null
  key_type    = "ed25519" -> null
  public_key  = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPNuu9F8koFKzEhsRwTKKLcwtL0kr5yY8EtDI/aQ+THD_codespace/codespaces-79c519" -> null
  region      = "me-central-1" -> null
  tags        = {} -> null
  tags_all    = {} -> null
}
```

```
# aws_subnet.myapp_subnet_1 will be destroyed
resource "aws_subnet" "myapp_subnet_1" {
  arn                  = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-03e5bda232f8fa581" -> null
  assign_ipv6_address_on_creation = false -> null
  availability_zone     = "me-central-1a" -> null
  availability_zone_id  = "mecl-az1" -> null
  cidr_block            = "10.0.10.0/24" -> null
  enable_dns64          = false -> null
  enable_inj_at_device_index = 0 -> null
  enable_resource_name_dns_a_record_on_launch = false -> null
  enable_resource_name_dns_aaaa_record_on_launch = false -> null
  id                   = "subnet-03e5bda232f8fa581" -> null
  ipv6_native          = false -> null
  map_customer_owned_ip_on_launch = false -> null
  map_public_ip_on_launch = false -> null
  owner_id             = "623705167686" -> null
  private_dns_hostname_type_on_launch = "ip-name" -> null
  region               = "me-central-1" -> null
}
```

Plan: 0 to add, 0 to change, 7 to destroy.

```
Changes to Outputs:
- ec2_public_ip = "3.29.125.164" -> null
- subnet_id     = "subnet-03e5bda232f8fa581" -> null
- vpc_id        = "vpc-08c0de79b0e26507d" -> null
aws_default_route_table.main_rt: Destroying... [id=rtb-0e0e0308f83c69ff2]
aws_instance.myapp_server: Destroying... [id=i-097e723716367b59c]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-08deab4a9bafbaac7]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-08deab4a9bafbaac7, 00m10s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-08deab4a9bafbaac7, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 27s
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m30s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m40s elapsed]
aws_instance.myapp_server: Destruction complete after 40s
aws_key_pair.ssh_key: Destroying... [id=serverkey]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-03e5bda232f8fa581]
aws_default_security_group.myapp_sg: Destroying... [id=sg-0a106f65daa2ea6eb]
aws_default_security_group.myapp_sg: Destruction complete after 0s
aws_key_pair.ssh_key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-08c0de79b0e26507d]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 7 destroyed.

@23-22412-038-sys → /workspaces/Lab11 (main) \$ |